4.Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

CODE:

```
clc;
clear;
close all;
%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');
```

```
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```

COMMAND WINDOW:

```
--- Rectangular Window Coefficients ---
 0.0126
 0
 -0.0137
 0
 0.0150
 0
 -0.0166
 0
 0.0185
 0
 -0.0210
 0
 0.0242
 0
 -0.0286
 0
 0.0349
 0
 -0.0449

 0
 0.0629
 0
 -0.1048
 0
 0.3145
 0.4940
 0.3145
```

0
-0.1048
0
0.0629
0
-0.0449
0
0.0349
0
-0.0286
0
0.0242
0
-0.0210
0
0.0185
0
-0.0166
0

0.0150
0
-0.0137
0
0.0126

--- Hamming Window Coefficients ---
0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125

0

-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0

-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010

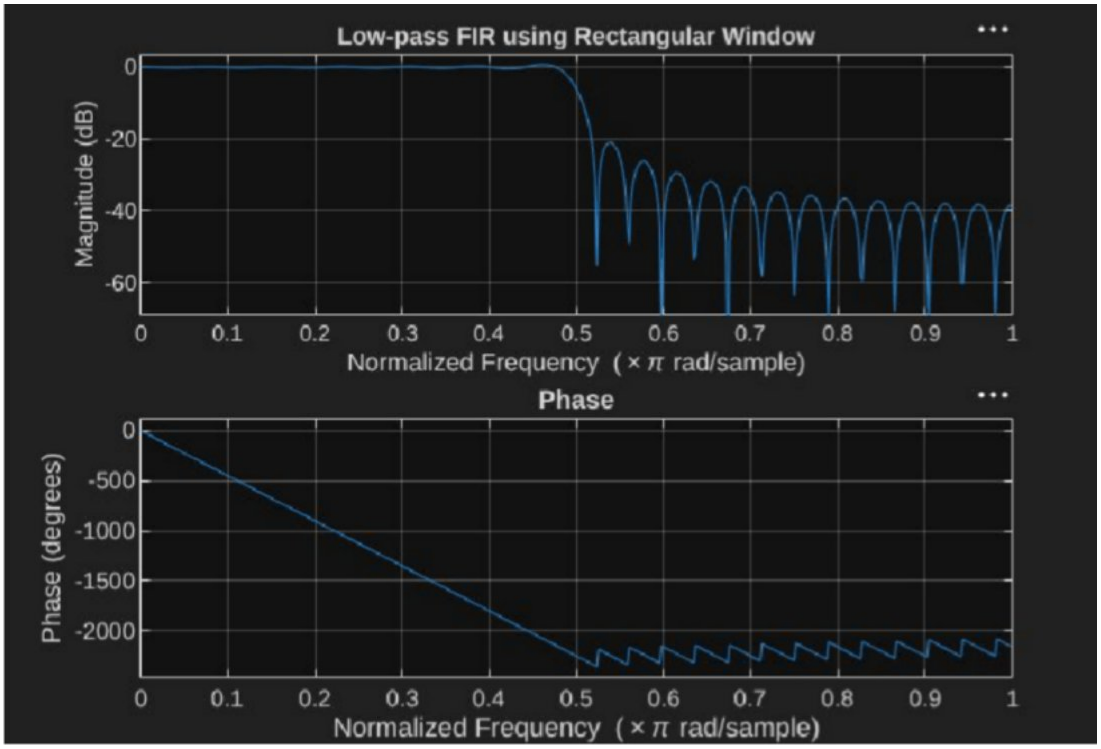
--- Hanning Window Coefficients ---
0.0000
0
-0.0004
0

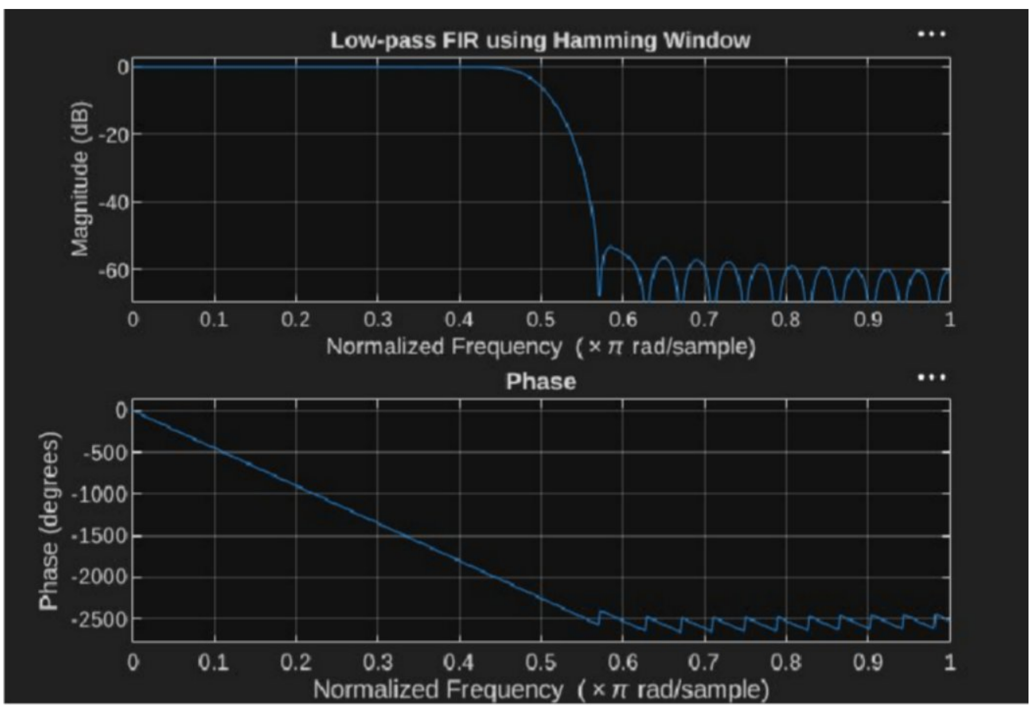
0.0013
0

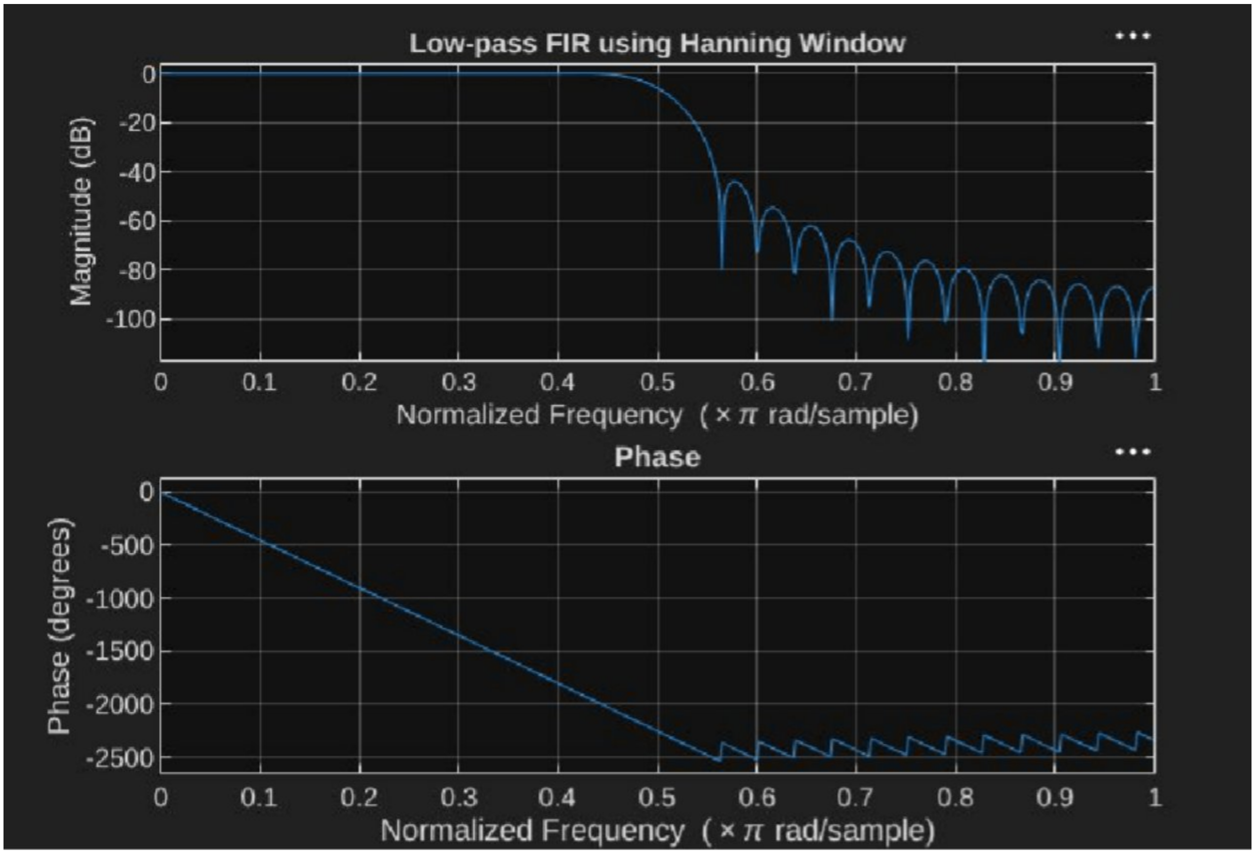
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0

-0.0179
0
0.0259
0
-0.0378
0
0.0580
0
-0.1027
0
0.3172
0.5000
0.3172
0
-0.1027
0
0.0580
0
-0.0378
0
0.0259
0
-0.0179
0
0.0122
0
-0.0081
0
0.0050
0
-0.0028
0
0.0013
0
-0.0004
0
0.0000

PLOTS:


Low-pass FIR using Rectangular Window
Magnitude (dB)
0
-20
-40
-60
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)


Low-pass FIR using Hamming Window
Magnitude (dB)
0
-20
-40
-60
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)



4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

CODE:

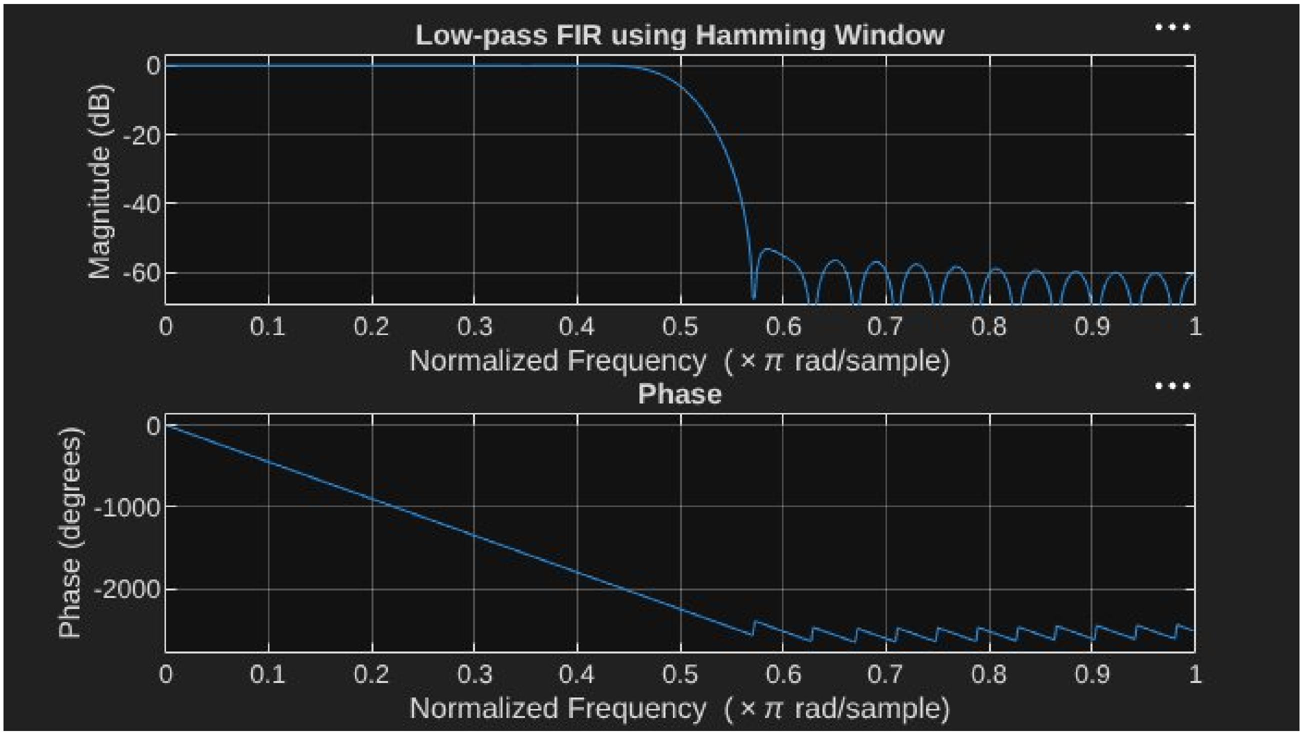
```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- Plot Frequency Responses ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```

Command Window:

```
--- Hamming Window Coefficients ---
    0.0010
         0
   -0.0013
         0
    0.0021
         0
   -0.0034
         0
    0.0055
         0
   -0.0084
         0
    0.0125
         0
   -0.0181
         0
    0.0260
         0
   -0.0379
         0
    0.0580
         0
   -0.1026
         0
    0.3168
    0.4995
    0.3168
         0
   -0.1026
         0
    0.0580
         0
   -0.0379
         0
    0.0260
         0
   -0.0181
         0
    0.0125
         0
   -0.0084
         0
    0.0055
         0
   -0.0034
         0
    0.0021
         0
```

```
   -0.0013
         0
    0.0010
>>
```

PLOTS:



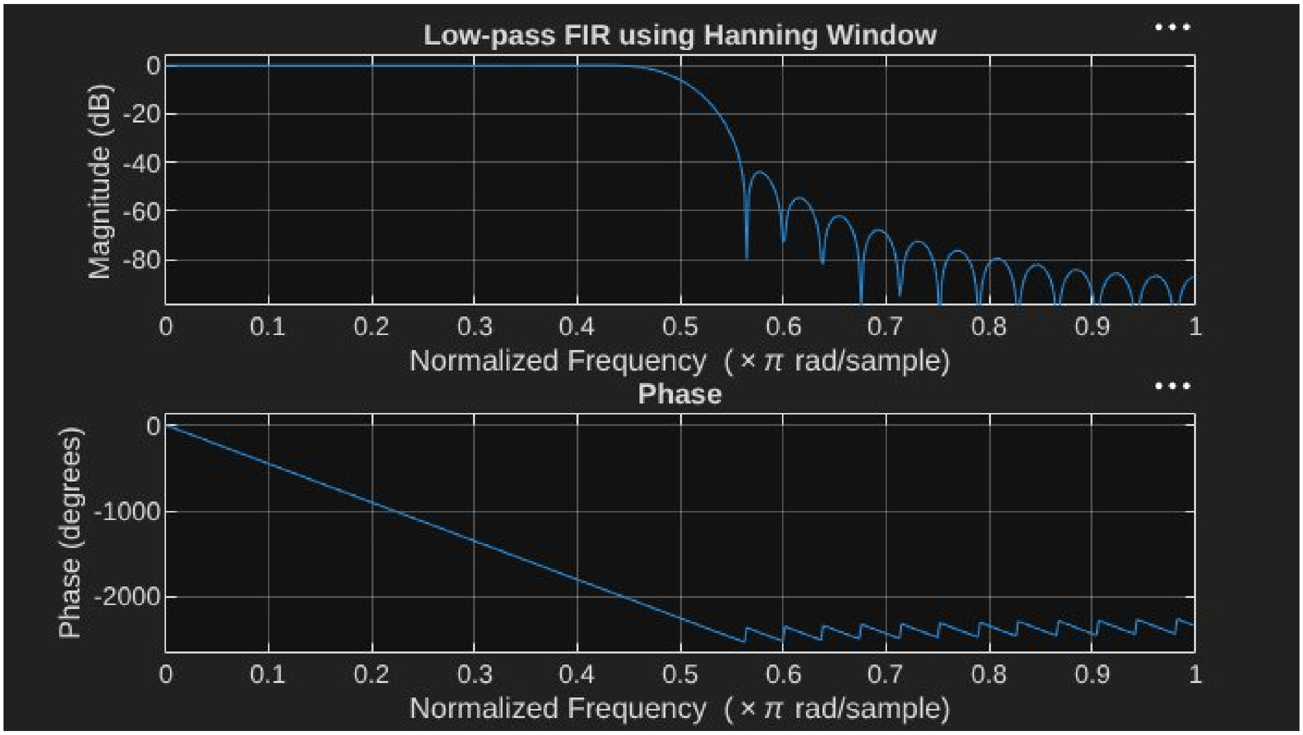
4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

Program (Matlab Code):

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

PLOT:



Low-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
-80
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)